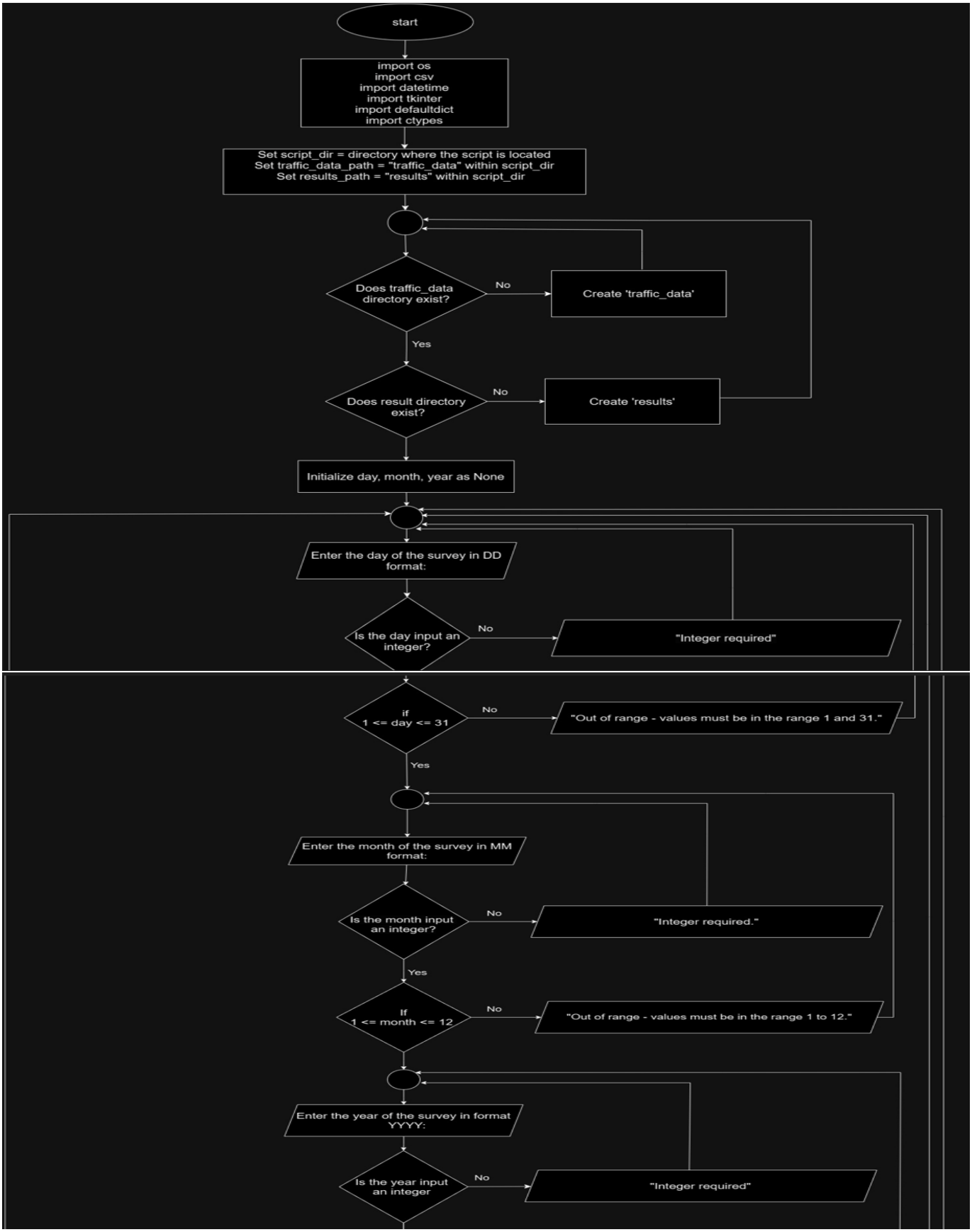
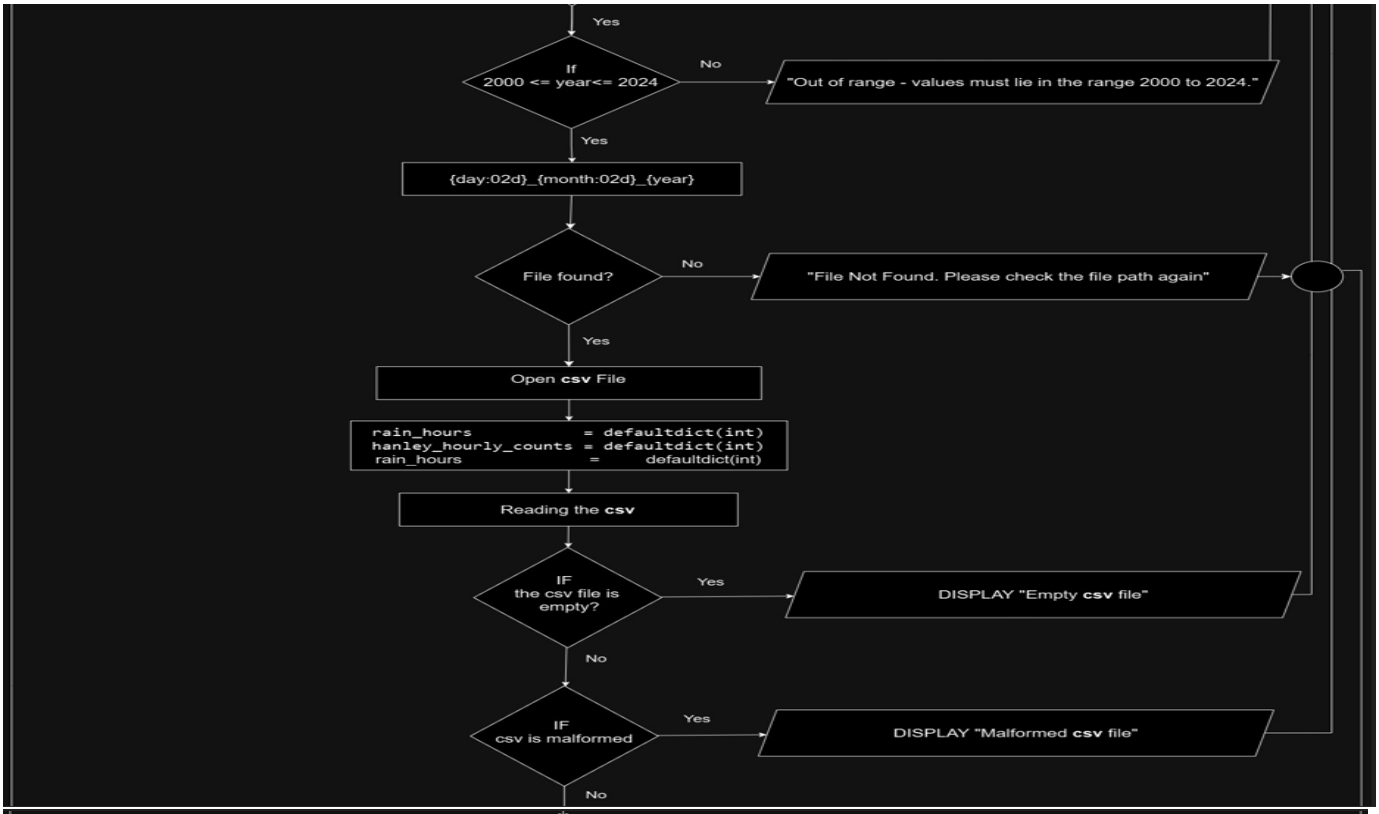
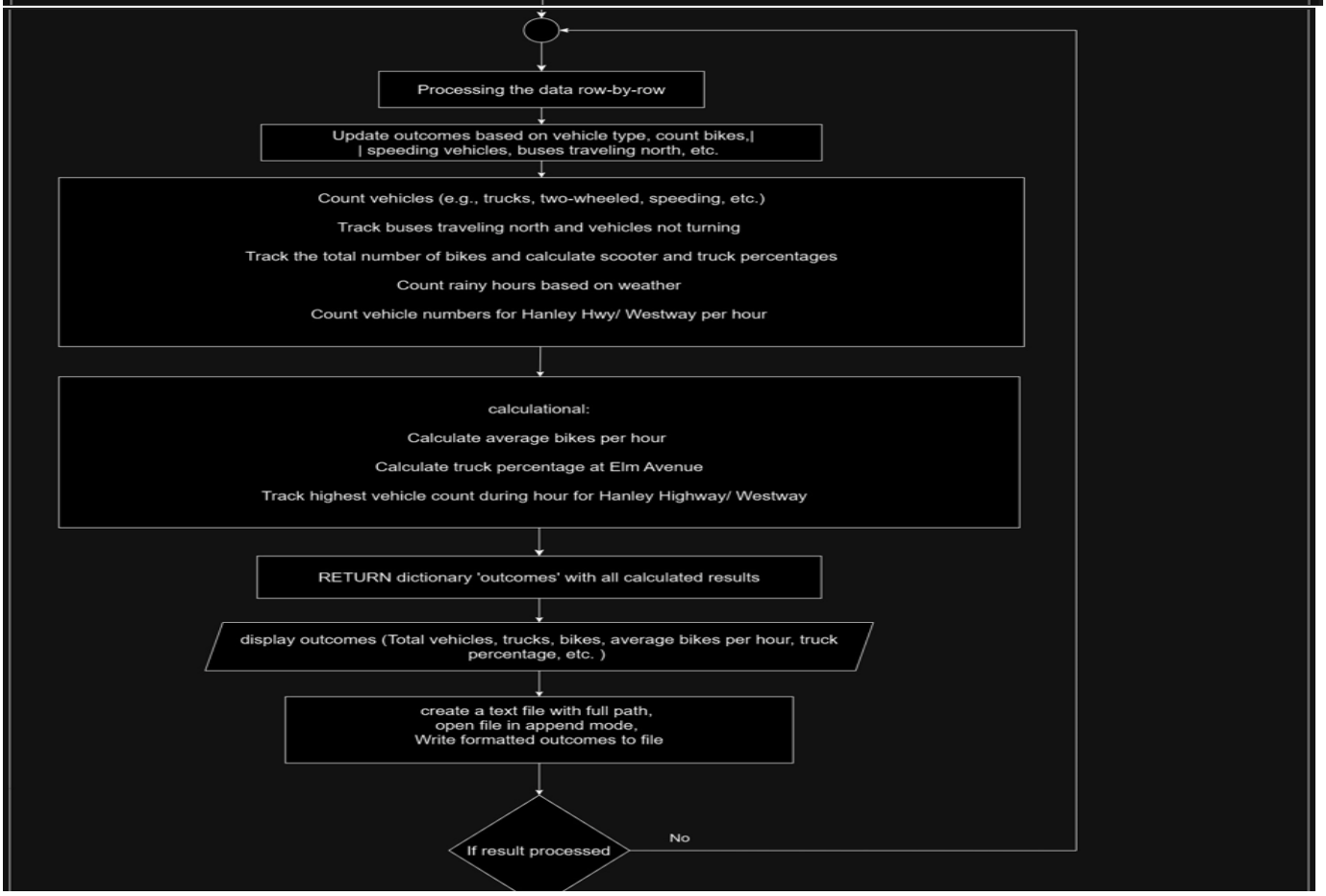
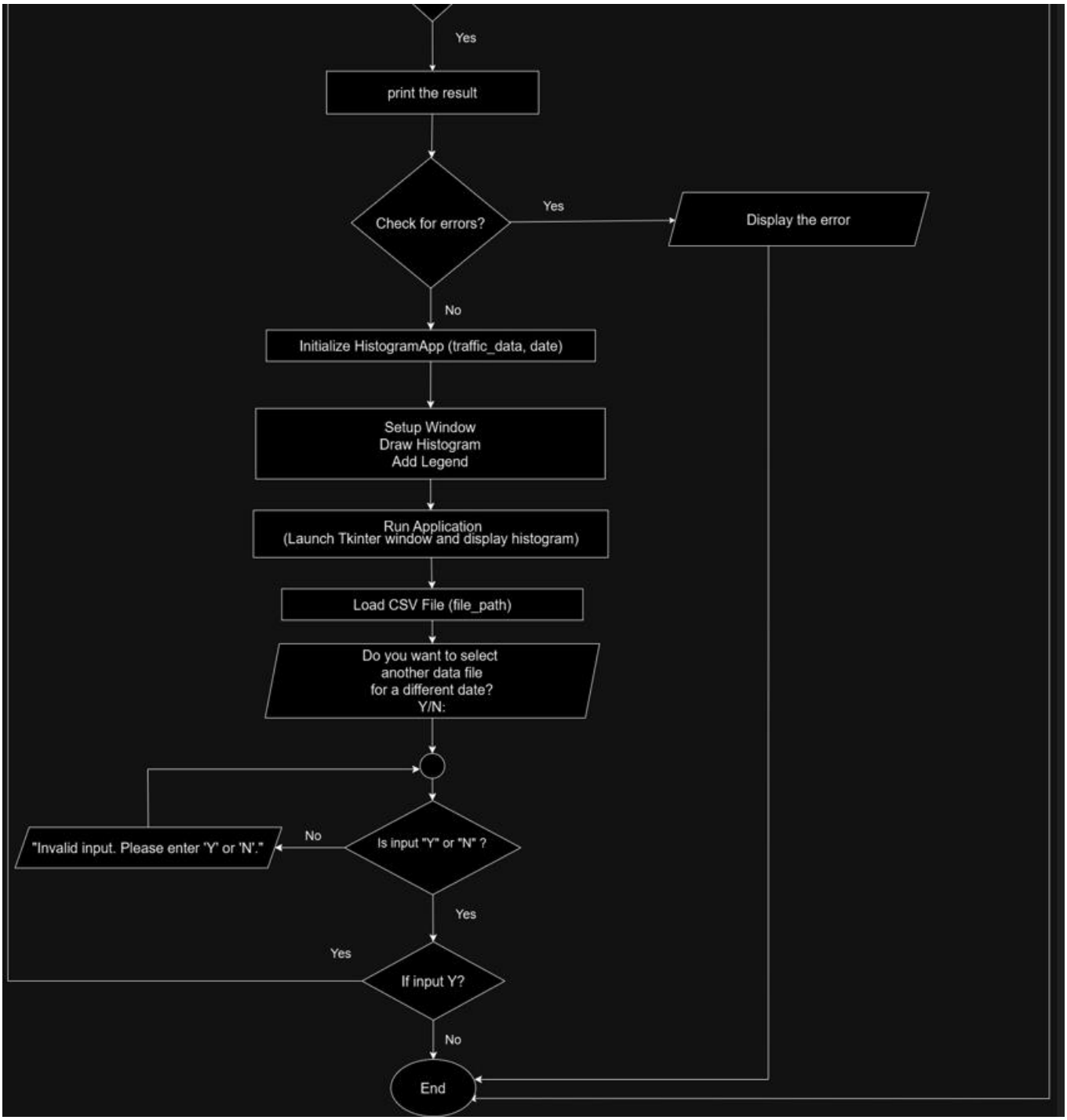

start
import os
import csv
import datetime
import tkinter
import defaultdict
import ctypes
Set script_dir = directory where the script is located
Set traffic_data_path = "traffic_data" within script_dir
Set results_path = "results" within script_dir
Does traffic_data directory exist?
No
Create 'traffic_data'
Yes
Does result directory exist?
No
Create 'results'
Initialize day, month, year as None
Enter the day of the survey in DD format:
Is the day input an integer?
No
"Integer required"
if 1 <= day <= 31
No
"Out of range - values must be in the range 1 and 31."
Yes
Enter the month of the survey in MM format:
Is the month input an integer?
No
"Integer required."
Yes
If 1 <= month <= 12
No
"Out of range - values must be in the range 1 to 12."
Enter the year of the survey in format YYYY:
Is the year input an integer
No
"Integer required"


Yes
If
2000 <= year<= 2024
No
"Out of range - values must lie in the range 2000 to 2024."
Yes
{day:02d}_{month:02d}_{year}
File found?
No
"File Not Found. Please check the file path again"
Yes
Open csv File
rain_hours = defaultdict(int)
hanley_hourly_counts = defaultdict(int)
rain_hours = defaultdict(int)
Reading the csv
IF
the csv file is
empty?
Yes
DISPLAY "Empty csv file"
No
IF
csv is malformed
Yes
DISPLAY "Malformed csv file"
No


Processing the data row-by-row
Update outcomes based on vehicle type, count bikes,|
| speeding vehicles, buses traveling north, etc.
Count vehicles (e.g., trucks, two-wheeled, speeding, etc.)
Track buses traveling north and vehicles not turning
Track the total number of bikes and calculate scooter and truck percentages
Count rainy hours based on weather
Count vehicle numbers for Hanley Hwy/ Westway per hour
calculational:
Calculate average bikes per hour
Calculate truck percentage at Elm Avenue
Track highest vehicle count during hour for Hanley Highway/ Westway
RETURN dictionary 'outcomes' with all calculated results
display outcomes (Total vehicles, trucks, bikes, average bikes per hour, truck
percentage, etc.)
create a text file with full path,
open file in append mode,
Write formatted outcomes to file
If result processed
No


Yes
print the result
Check for errors?
Yes
Display the error
No
Initialize HistogramApp (traffic_data, date)
Setup Window
Draw Histogram
Add Legend
Run Application
(Launch Tkinter window and display histogram)
Load CSV File (file_path)
Do you want to select
another data file
for a different date?
Y/N:
"Invalid input. Please enter 'Y' or 'N'."
No
Is input "Y" or "N" ?
Yes
Yes
If input Y?
No
End